

Physics Class 11 Federal Board Solved MCQs of Model Paper and All Past Papers of 2025 with Explanation (Very Important)

i.	Which of the following quantities are dimensionally similar?	Torque and work	Stress and spring constant	Angle and length	Strain and pressure
ii.	Maximum work is obtained by the process called:	Isochoric	Isothermal	Isobaric	Adiabatic
iii.	Projectile motion is two-dimensional motion under constant acceleration due to	Air resistance	Gravity	Horizontal distance	Velocity

a) $\tau = rF = Nm$
 $W = Fd = Nm$
 $\text{Stress} = p = \frac{F}{A} = Nm^{-2}$
 $K = \frac{F}{x} = \frac{N}{m} = Nm^{-1}$
 $l = r\theta = Q = \frac{l}{r}$

Strain = $\frac{\Delta L}{L}$
 $\Delta V = 0$ $\Delta T = 0$ $\Delta p = 0$ $\Delta Q = 0$

ii.	Maximum work is obtained by the process called:	Isochoric	Isothermal	Isobaric	Adiabatic
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1st Law of Thermodynamics

$$Q = \Delta U + W \rightarrow (1)$$

a) $W = P\Delta V = 0$

b) $\Delta T = 0 \Rightarrow \Delta U = 0 \Rightarrow (1) \Rightarrow \boxed{Q = W}$

c) $\Delta p = 0 \Rightarrow W = Q - \Delta U \Rightarrow W < Q$

$x = 5, y = 2$

$$y = x - 3, \quad x=5, y=2$$

$$d) \quad 0 = \Delta U + W \Rightarrow \boxed{W = -\Delta U}$$

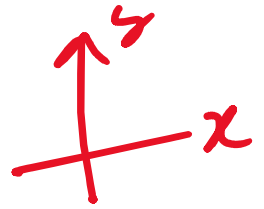
$$iii) \quad \boxed{a_y = -g} \quad \boxed{a_x = 0 \text{ m/s}^2}$$

$$a = \frac{dv}{dx}$$

$$v_x = \text{const}$$

$$v_y$$

$$v_x$$



$$1 \text{ h} = 60 \text{ min}$$

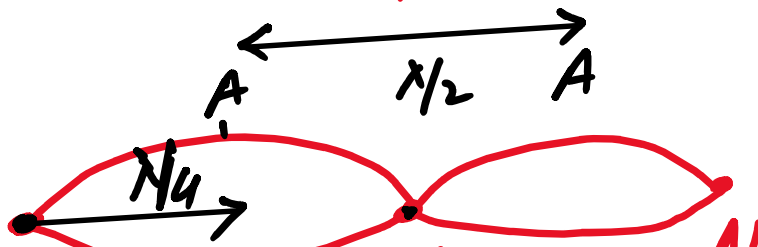
$$1 \text{ mi} = 60 \text{ s}$$

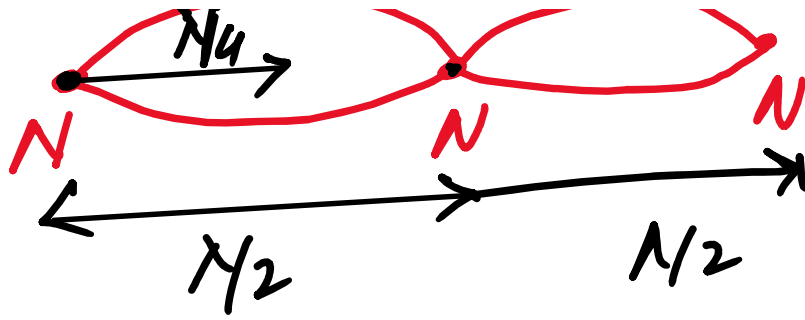
iv.	A turtle travel 45cm in 20 minutes, its average velocity will be:	2.25cm/h	90cm/h	135cm/h	15cm/h
v.	A stationary wave is formed by waves of frequency 512Hz. The speed of the wave is 65m/s, the distance between two consecutive antinodes is:	0.126m	0.06m	0.03m	0.024
vi.	A wire is stretched to double of its original length, its strain is	0.5	1	zero	2

$$t = 20 \text{ min} = \frac{20}{60} \text{ h} = \frac{1}{3} \text{ h}$$

$$v = \frac{45 \text{ cm}}{\frac{1}{3} \text{ h}} = 135 \text{ cm/h}$$

$$v) \quad f = 512 \text{ Hz}, \quad v = 65 \text{ m/s}$$





$$v = f\lambda \Rightarrow \lambda = \frac{v}{f} = \frac{65 \text{ m/s}}{512 \text{ s}^{-1}}$$

$$d = \frac{\lambda}{2} = 0.0634 \text{ m}$$

$$= 0.06 \text{ m}$$

Strain $\frac{\Delta L}{L} \Rightarrow$

$$L \rightarrow 2L$$

$$\Delta L = 2L - L = L$$

$$= \frac{L}{L} = 1$$

$$L \rightarrow \frac{L}{2} \Rightarrow L - \frac{L}{2} = \frac{L}{2}$$

$$\text{Strain} = \frac{\Delta L}{L} = \frac{\frac{L}{2}}{L} = \frac{1}{2}$$

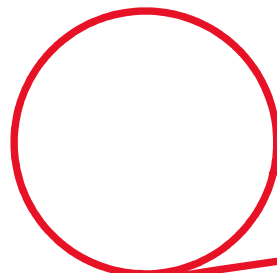
$$I = 0$$

vii.	Cyclotron frequency of a charged particle moving in a magnetic field B is independent of:	Magnetic flux density	Charge on a particle	Velocity of particle	Mass of particle
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vii.	Cyclotron frequency of a charged particle moving in a magnetic field B is independent of:	Magnetic flux density	Charge on a particle	Velocity of particle	Mass of particle
viii.	The best instrument for accurate measurement of <u>emf</u> of a cell is:	Voltmeter	Potentiometer ✓✓	Ammeter	Ohmmeter
ix.	What is the base unit of pressure?	kgms ⁻¹	kgm ⁻¹ s ⁻¹	kgm ⁻¹ s ⁻²	kgm ² s ⁻²
x.	Theory of relativity was formulated by.	Isaac Newton	Rutherford	Plank	Albert Einstein

$$F = qvB$$

$$F = \frac{mv^2}{r}$$

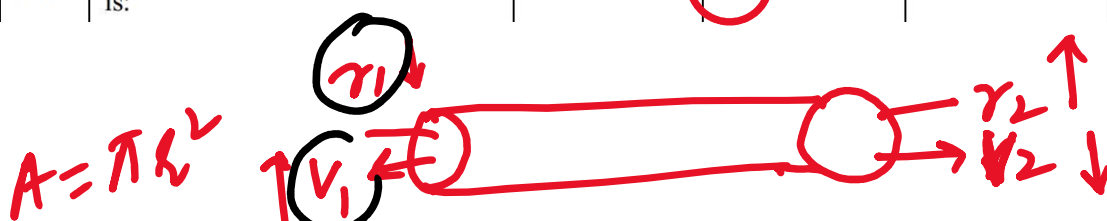


$$T = \frac{2\pi r}{v} \quad v = \frac{s}{t}$$

$$f = \frac{1}{T} = \frac{qB}{2\pi m}$$

$$P = \frac{F}{A} = \frac{ma}{L^2} = \frac{\text{kg ms}^{-2}}{\text{m}^2} = \text{kg m}^{-1} \text{s}^{-2}$$

xi.	The radius at two ends of a pipe is in the ratio of 2:3, then the speed of liquid at the two ends is in the ratio of:	2:3	3:2	4:9	9:4 ✓
xii.	Lepton number for muon particle is:	-1	+1	+1/2	0



$$A = \pi R^2 \quad \uparrow (V_1, E) \quad \rightarrow \quad V_2 \downarrow$$

$$A_1 V_1 = A_2 V_2 = \text{const}$$

$$v \propto \frac{1}{A}$$

$$\frac{r_1}{r_2} = \frac{2}{3}$$

$$\frac{V_1}{V_2} = ?$$

$$\frac{A_1}{A_2} = \frac{V_2}{V_1}$$

$$\frac{\pi R_1^2}{\pi R_2^2} = \frac{V_2}{V_1} \Rightarrow \frac{4}{9} = \frac{V_2}{V_1}$$

$$\Rightarrow \boxed{\frac{V_1}{V_2} = \frac{9}{4}}$$

xiv) Lepton & Quarks

$\left. \begin{array}{l} \text{electron}(e^-), \text{muon}(\mu^-) \\ \text{Tau}(\tau^-), \text{Neutrino}(\nu) \end{array} \right\} \begin{array}{l} \nu \\ (+1) \end{array}$

positron, Antineutrino ($\bar{\nu}$),
Antimatter } -1

Strong
Nuclear
force $\leftarrow$ Quarks ✓
↓
Hadrons

Lepton $\rightarrow$ Do not
experience
nuclear

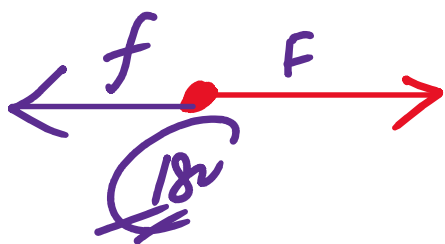
Hadrons
↓
Baryons (3 quarks)
↓
Mesons (2 quarks)

Proton, Neutron
π (pions) + Kaon (K)

Leptons
↓
electrons, neutrinos ✓

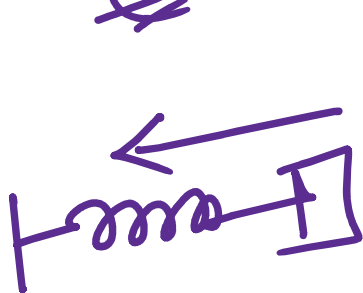
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xiii.	Which of the following moving particle is not deflected by magnetic field?	Neutron	Proton	Electron	Positron
xiv.	The force that acts on the body but it does no work is called:	Frictional force	Gravitational force	Centripetal force	Elastic force
xv.	The region around a charge in which a test charge can feel an electric force is called	Electric intensity	Electric Potential	Electric Field	Electric Energy
xvi.	Moment of inertia of a spinning body about its axis does not depend on the:	Angular velocity of body	Mass of body	Orientation of the axis	Distribution of the mass around axis
xvii.	When $F_x = 3 \text{ N}$ and $F = 5 \text{ N}$ then $F_y =$	6N	8N	2N	4N



$$W = F_s \cos \theta$$

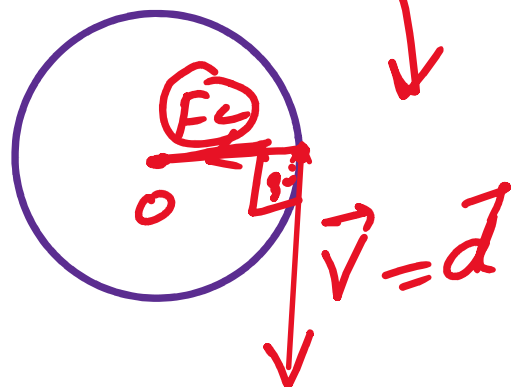
$$V = \frac{W}{q}$$



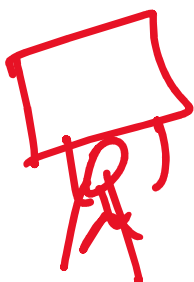
$$E = mgh$$

$$W = F_s \cos \theta$$

$$v = \frac{d}{t}$$



$$W = F_c d \cos \theta$$



2

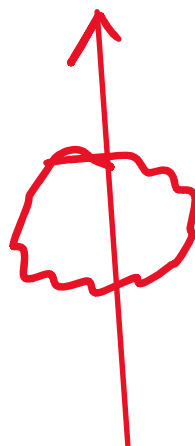
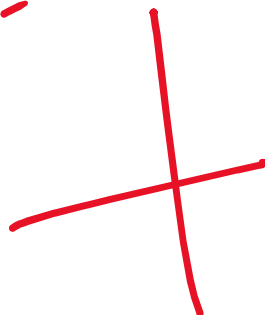


ii)

$$I = \frac{mr^2}{2}$$

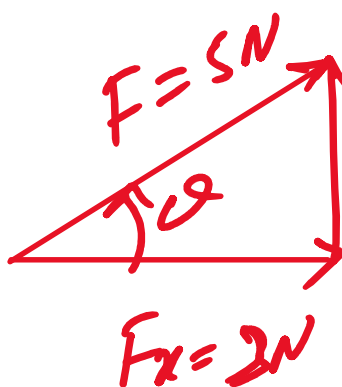
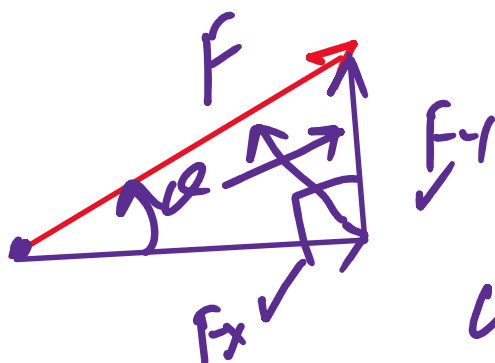


3, 4, 5



$$25 = 9 + 16$$

iii)



$$F_y = 2$$

$$= 4N$$

$$25 = 9 + F_y^2$$

$$\cos \theta = \frac{F_x}{F} \Rightarrow F_x = F \cos \theta$$

3, 4, 5



6, 8, 10

18, 12, 11

5, 12, 13

10, 24, 26

$$169 = 25 + 144$$

101

Physics MCQs Local Area 1st Annual 2025

$[LT^{-1}]$ $[LT^{-2}]$

1. The dimensions $[ML^2T^{-2}]$ show:	Power	Work	Velocity	Acceleration
2. If $r = 2.25 \pm 0.01 \text{ cm}$ then percentage uncertainty in "r" will be:	0.2%	0.3%	0.4%	0.5%
3. $\hat{j} \cdot (\hat{i} \times \hat{k}) =$	0	-1	1	2
4. Magnitude of centripetal force $F_c =$	$m\omega^2$	$m\omega$	$n\omega^2$	mv^2
5. If the length of blade of a fan is increased, then its rotational speed (at same voltage and current):	Will increase	Will decrease	Will become infinite	Remains constant
6. Two bodies A and B of same masses have velocities in the ratio 2:1, their kinetic energies are in the ratio:	2:1	1:2	4:1	1:4
7. Maximum drag force on a 1kg falling sphere is:	9.8 N	1 N	98 N	4.9 N
8. Young's Modulus, $Y =$	$\frac{\Delta L / L}{F / A}$	$\frac{FA}{\Delta L L}$	$\frac{\Delta L L}{FA}$	$\frac{F / A}{\Delta L / L}$
9. In an adiabatic process which quantity remains constant?	Internal energy	Volume	Temperature	Mass
10. Location of a submarine in deep sea can be detected by:	Doppler effect	Compton's effect	Faraday's Law	Bernoulli principle

$$= \frac{0.01}{2.25} \times 100\%$$

$$\frac{F/A}{\Delta L / L}$$

$$dQ = 0 \quad Q = W + \Delta U$$

$$E = mc^2 \quad P = \frac{W}{T}$$

SONAR

$$W = mgh = \text{maxd}$$

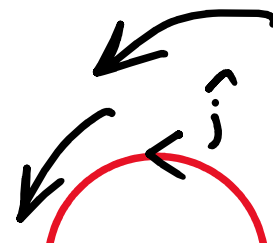
$$\text{kg} \cdot \text{m} \cdot \text{s}^{-2} \times \text{m}$$

$$\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-2} \quad [ML^2T^{-2}]$$

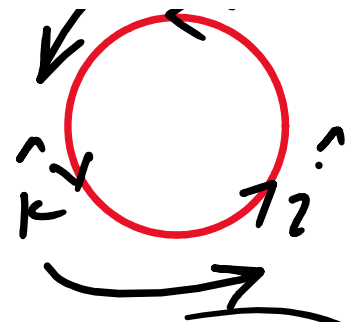
$$P = \frac{ML^2T^{-2}}{T}$$

$$= ML^2T^{-3}$$

$$\hat{j} \Rightarrow \hat{i} \times \hat{k} = \hat{j} \cdot (-\hat{i})$$



$$iii) j \cdot \underline{(i \times k)} = j \cdot (-1)$$



$$\boxed{i \times i = 0}$$

$$i \times j = k$$

$$j \times k = i$$

$$k \times i = j$$

$$\Rightarrow i \times k = -j$$

$$\boxed{i \cdot i = j \cdot j = k \cdot k = 1}$$



$$i \cdot j = i \cdot k = j \cdot k = 0$$

$$\cos 90^\circ = 0$$

$$v = r\omega$$

$$iii) F_c = \frac{mv^2}{r} = \frac{m(r^2\omega^2)}{r}$$

$$= m r \omega^2$$

$$\downarrow$$

$$L = \underline{I\omega}$$

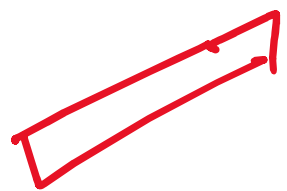
$$\omega = \frac{L}{I} \Rightarrow$$

$$\omega \propto \frac{1}{I}$$

$$m \uparrow \checkmark$$

$$I \uparrow$$

$$I = m r^2$$



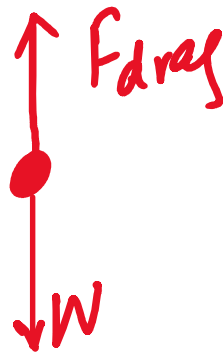
$$\frac{V_1}{V_2} = \frac{2}{1}$$

$$\frac{K_1}{K_2} = ?$$

$$\frac{K_1}{K_2} = \frac{\frac{1}{2} m V_1^2}{\frac{1}{2} m V_2^2} = \left(\frac{V_1}{V_2} \right)^2 = \left(\frac{2}{1} \right)^2 = \frac{4}{1}$$

$$\Rightarrow 4:1$$

7)



$$F_d = mg$$

$$= 1 \text{ kg} (9.8)$$

$$F_d = 9.8 \text{ N}$$

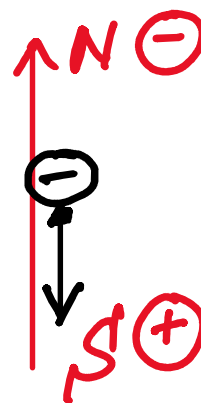
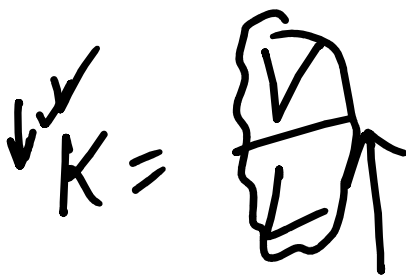
$$Y = \frac{F}{A} \div \frac{\Delta L}{L}$$

$$= \frac{F}{A} \times \frac{L}{\Delta L} \Rightarrow$$

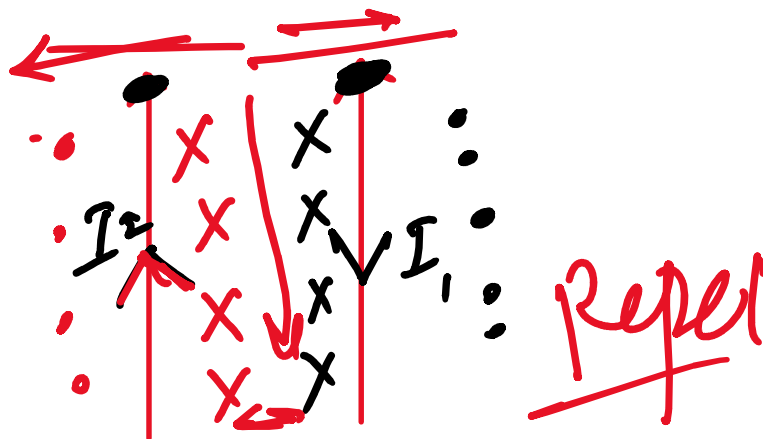
$$Y = \frac{FL}{A \Delta L}$$

11. An electron is placed in a uniform electric field directed from south to north. In which direction will electron move after it is released?	East	South	North	West
12. In order to achieve high accuracy, the slide wire of a potentiometer should be:	As long as possible	As short as possible	Very thick	Neither too small, nor too large
13. Two closely placed parallel wires are carrying current in opposite direction, they will:	Repel each other	Attract each other	Not affect each other	Rotate each other
14. One weber is equal to:	NA^{-1}	NmA^{-1}	$Nm^{-1}A$	$Nm^{-1}A^{-1}$
15. When an object moves with a very high speed its length _____ in the direction of its motion.	Contracts	Expands	Remains same	Becomes zero
16. Relativistic mass is always:	Less than rest mass	Greater than rest mass	Equal to rest mass	Zero
17. A hadron may either be a Baryon or a:	Lepton	Gluon	Photon	Meson

2 quarks
 $\pi + K$



RHR



~~Attraction~~ Attraction

$$14) \quad \Phi_B = B_A \Rightarrow N \bar{A}^{-1} m^{-1} \times m^2 = \bar{N} \bar{A}^{-1} m$$

$$F = ILB \Rightarrow B = \frac{F}{IL} = \frac{N}{Am}$$

$$= \bar{N} \bar{A}^{-1} m^{-1}$$

$$y = 0.5x \quad x=1$$

$$L = L_0 \sqrt{1 - \frac{v^2}{c^2}} \quad v = 0.9c$$

$$L < L_0$$

$$v \sim c$$

$$\sqrt{1 - 0.9} = \sqrt{0.1}$$

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$v^2 = 0.9c^2$$

$$\sqrt{1 - 0.9} = \sqrt{0.1}$$

Phy 1st Annual Hold Area 2021

→ 2

1.	In a simple pendulum experiment, percentage errors (uncertainty) in "L" and "T" are 0.1% and 2% respectively. Maximum percentage error (uncertainty) in "LT ² " is:	4.1%	3%	2.1%	3.2
2.	If $\vec{A} \cdot \vec{B} = \vec{A} \times \vec{B} $ then angle "θ" between vectors $\vec{A}$ and $\vec{B}$ is:	0	$\pi/4$	$\pi/2$	π
3.	When a massive body of mass M_1 is moving with velocity V_1 collides with lighter stationary body of mass m_2 , the velocity of massive body after collision will be:	$V'_1 = 2V_1$	$V'_2 = V_1$	$V'_1 = V_1$	$V'_2 = 2V_2$
4.	Moment of inertia of a body spinning about its axis DOES NOT depend on the:	Angular velocity of body	Mass of body	Orientation of the axis	Distribution of mass around axis
5.	The bodies A and B of equal masses have velocities ratio in 2:1 their K.E are in the ratio of:	2:1	1:2	4:1	1:4
6.	The fluid is said to be incompressible, if its density is:	Constant	Very high	Very small	Zero
7.	Equation of continuity is based upon law of conservation of:	Momentum	Energy	Mass ✓	Charge
8.	Glass is an example of _____ solid.	Crystalline	Amorphous	Poly-crystalline	Ductile
9.	Young's Modulus, Y=	$\frac{\Delta L / L}{F / A}$	$\frac{FA}{\Delta L L}$	$\frac{\Delta L L}{FA}$	$\frac{F / A}{\Delta L / L}$
10.	A Carnot engine undergoes _____ processes in one cycle.	Two	Three	Four	Six

$$0.1\% + 2(2\%) = 4.1\%$$

Answer
AB Since

$$\frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$



3) $M_1 \gg m_2$
 V_1
 0

iso exp
Adapt
iso temp
Adiabatic

$$V'_1 = \left(\frac{M_1 - m_2}{M_1 + m_2} \right) V_1 = \left(\frac{M_1}{M_1} \right) V_1$$

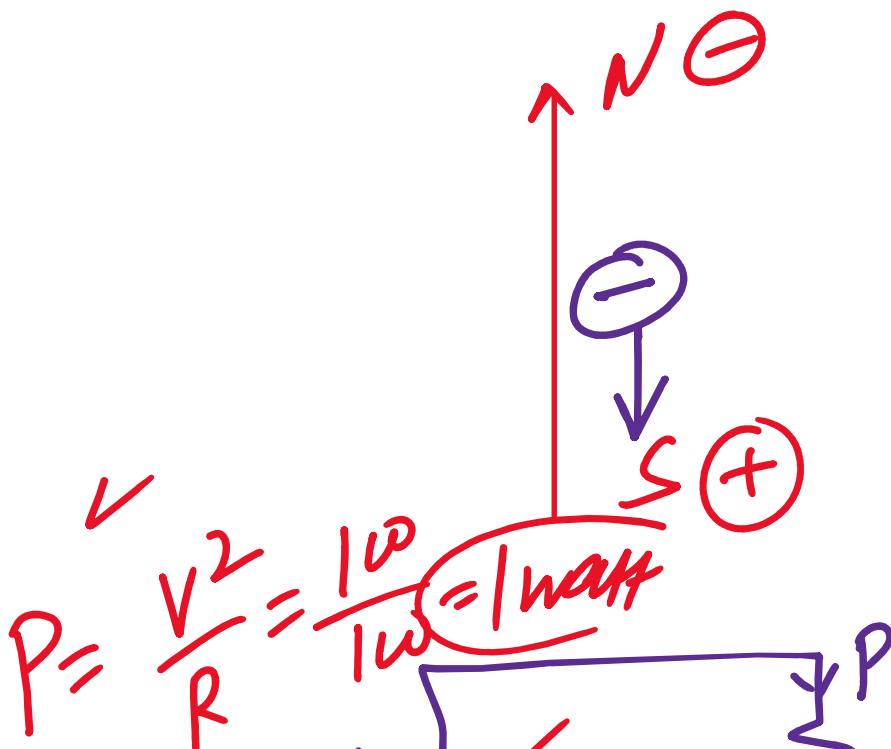
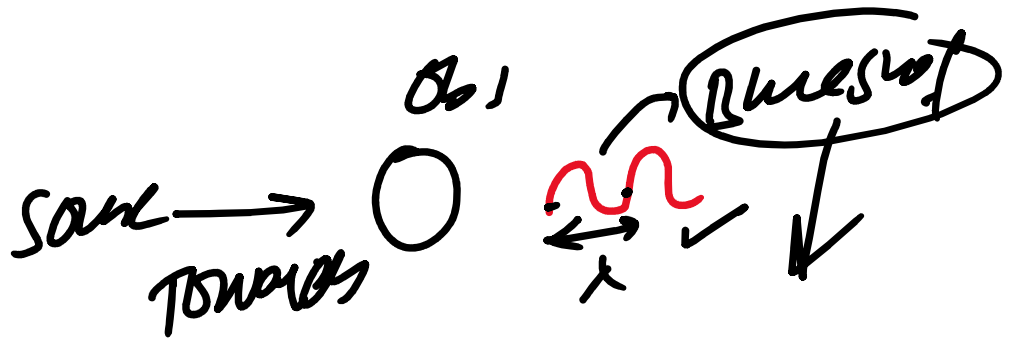
$$M_1 \gg m_2 \Rightarrow M_1 - m_2 = M_1$$

$$M_1 + m_2 = M_1$$

11. Stars moving away from the earth show:	Red shift	Blue shift	Black shift	Yellow shift	☐	☐	☐
12. A negatively charged particle is placed in a uniform electric field directed from south to north. In which direction will the particle move after being released?	North	South	East	West	☐	☐	☐
13. A 100 ohm resistor is connected across the terminals of a 10V battery. The power dissipation of the resistor is:	1 watt	10 watts	100 watts	1000 watts	☐	☐	☐
14. The SI unit of magnetic induction is:	Nm	Gauss	Weber	Tesla	☐	☐	☐
15. Relativistic mass is always:	Less than rest mass	Greater than rest mass	Equal to rest mass	Zero	☐	☐	☐
16. Inertial frame of reference is a frame in which law of motion holds.	1st	2nd	3rd	4th	☐	☐	☐
17. Two up quarks and one down quark make a:	Meson	Photon	Neutron	Proton	☐	☐	☐

1HA-I 25004 (D)

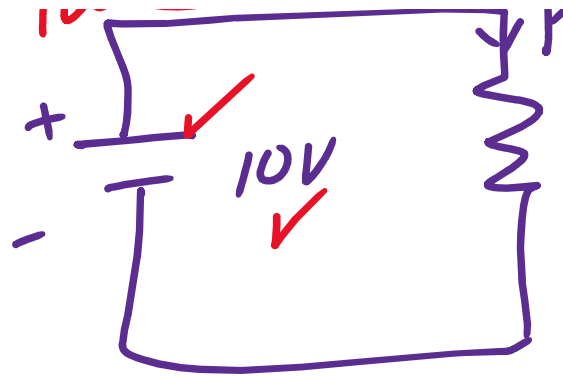
$(K.E)_{max} = \frac{1}{2}mv^2$ • $l = mr^2$ • $P = VI$ • $P = \frac{V^2}{R}$ • $\vec{A} \cdot \vec{B} = AB \cos \theta$ • $\vec{A} \times \vec{B} = AB \sin \theta$ • $L = I\omega$ • $v = r\omega$



$P = I^2 R$

$P = VI$

$$V = IR$$



$$P = VI$$

$$U U d$$

$$+\frac{2}{3}e + \frac{2}{3}e + (-\frac{1}{3}e)$$

$$= \frac{3}{3}e = (+1e) \rightarrow \text{net } +$$

Phy. 2nd Annual Exams 2025 MCQs

1. The unit of force in terms of base units:	kg·m/s	kg·m/s ² ✓	kg·m ² /s	kg ² ·m/s ²
2. The dimension [ML ⁻¹ T ⁻²] represents:	Pressure ✓	Force	Power	Work
3. Torque $\vec{\tau}$ on a body is given by:	$\vec{\tau} = \vec{m} \times \vec{a}$	$\vec{\tau} = \vec{F} \times \vec{d}$	$\vec{\tau} = \vec{r} \times \vec{F}$ ✓	$\vec{\tau} = \vec{l} \times \vec{\omega}$
4. For a freely falling body, the vertical velocity at the 4 th second is:	94.9 ms ⁻¹	47.5 ms ⁻¹	40 ms ⁻¹	39.2 ms ⁻¹ ✓
5. The maximum range of a projectile is obtained when launch angle $\theta =$	30°	45° ✓	60°	90°
6. The SI unit of angular velocity is:	Rad	N·m	m/s ²	rad/s ✓
7. The centripetal force for a body of mass m moving with speed v in a circle of radius r is:	mv	mv ² r	$\frac{mv^2}{r}$ ✓	$\frac{m}{r^2}$
8. The work done by a gas at constant pressure is given by:	P·ΔV ✓	$\frac{\Delta V}{P}$	ΔV	V·ΔP
9. The first law of thermodynamics is expressed as:	Q = U + V	Q = T + S	Q = ΔU + W ✓	Q = mL
10. The Doppler effect explains the change in:	Frequency ✓	Amplitude	Speed of light	Wavelength only

$$F = ma$$

$$P = \frac{F}{A} = \frac{MLT^{-2}}{L^2} = ML^{-1}T^{-2}$$

$$\omega = \frac{\theta}{t}$$

$$V_i = 0 \text{ m/s}$$

$$v \cdot \text{rad}$$

$V_i = 0 \text{ m/s}$
 $V_f = V_i + at$
 $V_f = V_i + gt$
 $t = 4 \mu\text{s}$
 $V_f = gt$
 $= (9.8) \times 4$
 m/s^2

-1 A

$R = \frac{V_i^2 \sin^2 \theta}{g}$

$\sin 90^\circ = 1$
 $2\theta = 90^\circ$
 $\theta = 45^\circ$

$R_{\text{max}} = \frac{V_i^2}{g}$

$P = \frac{V^2}{R} = \frac{100}{10} = 10 \text{ W}$



11. For constructive interference, path difference Δx should be:	$n\lambda$	$\frac{(2n+1)\lambda}{2}$	$\frac{(n+1)\lambda}{2}$	$\frac{n\lambda}{2}$
12. Electric field intensity E is defined as:	F/d	F/q ✓	V/l	W/q
13. A 10Ω resistor is connected across the terminals of a 10V battery. The power dissipation of the resistor is:	100 W	50 W	10 W	25 W
14. Magnetic flux Φ through a surface area A is:	B/A	B·A ✓	A/B	B/A²
15. Lenz's law states that induced emf opposes:	Current	Resistance	Power factor	Change causing it
16. An inertial frame of reference is the one:	Which is at rest	Which has zero acceleration	Which is under a force	Which has variable acceleration
17. At what speed the mass of a body becomes double of its proper mass?	$\frac{2}{3}c$	$\frac{\sqrt{3}}{2}c$	$\frac{\sqrt{2}}{2}c$	$\frac{1}{2}c$

$\Delta x = n\lambda$
 $0, 1, 2, \dots$



16. An inertial frame of reference is the one:

	rest	zero acceleration	under a force	variable acceleration
17. At what speed the mass of a body becomes double of its proper mass?	$\frac{2}{3}c$	$\frac{\sqrt{3}}{2}c$	$\frac{\sqrt{2}}{3}c$	$\frac{1}{3}c$

1HA-II 25004

$$P = \frac{V^2}{R} = I^2 R = V \times I$$

$$g = 9.8 \text{ ms}^{-2}$$

$$F = ma$$

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$v_f = v_i + gt$$

Destructive
Dx =

$$(2n+1)\frac{\lambda}{2} \Rightarrow \frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}$$

$$= (n + \frac{1}{2})\lambda$$

$$17) \quad 2\gamma = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\Rightarrow 2 = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\Rightarrow 4 = \frac{1}{1 - \frac{v^2}{c^2}}$$

$$1 - \frac{v^2}{c^2} = \frac{1}{4}$$

$$\Rightarrow \frac{v^2}{c^2} = 1 - \frac{1}{4}$$

$$4 = 5$$

$$\frac{v^2}{c^2} = \frac{3}{4}$$

$$\Rightarrow \sqrt{v^2} = \sqrt{\frac{3}{4} c^2}$$

$$v = \frac{\sqrt{3}}{2} c$$